

Fire Alarm By Using Sensor



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Fire Alarm By Using Sensor

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Introduction

Smoke alarms detect fires by sensing small particles in the air using a couple of different kinds of technologies. Once they detect those particles above a certain level, they signal the alarm to sound so that you and your family can get to safety and call 911.

Photoelectric and Ionization Fire Alarm

Heavy smoke from a fire enters a chamber in the alarm with a LED light. The smoke will then deflect light onto a photoelectric sensor, activating the alarm. The complex circuitry and required chamber increases manufacturing price.

Overview

This project, we will build a Fire Detection System using Flame Sensor & Arduino. Earlier we build a simple Fire Detector Project using Thermistor. But this project uses a specific sensor called Flame Sensor instead of any temperature sensor.

Fire accidents can cause devastating damage to life and property. Early detection of fires is crucial for safety and prevention. With advancements in technology, it's possible to create a simple yet effective fire detection system using a flame sensor and an Arduino Code.

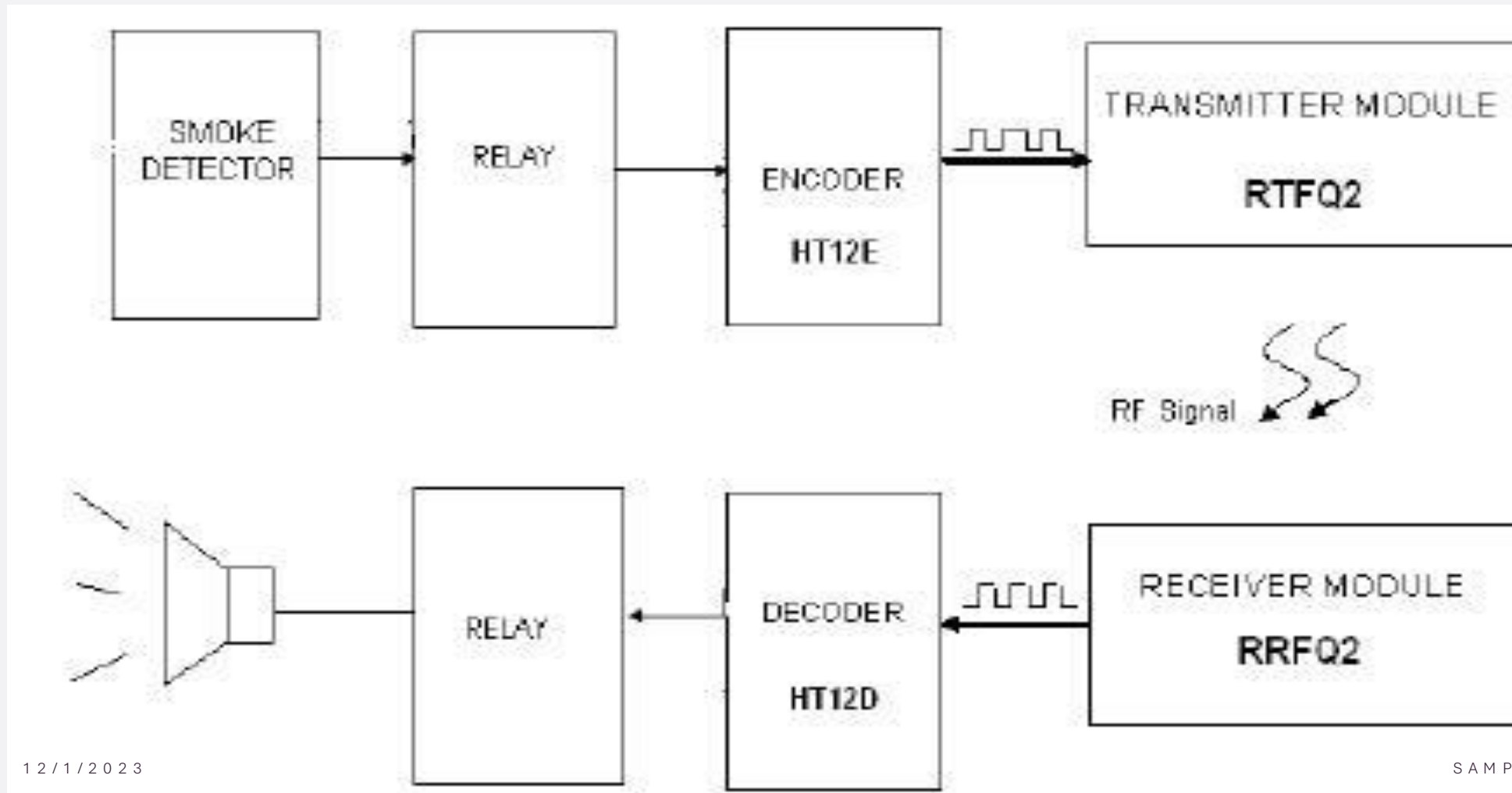
This post will discuss how to set up a fire detection system that not only senses the presence of a flame but also alerts the surroundings with a visual signal and a melody alarm.

Components Used

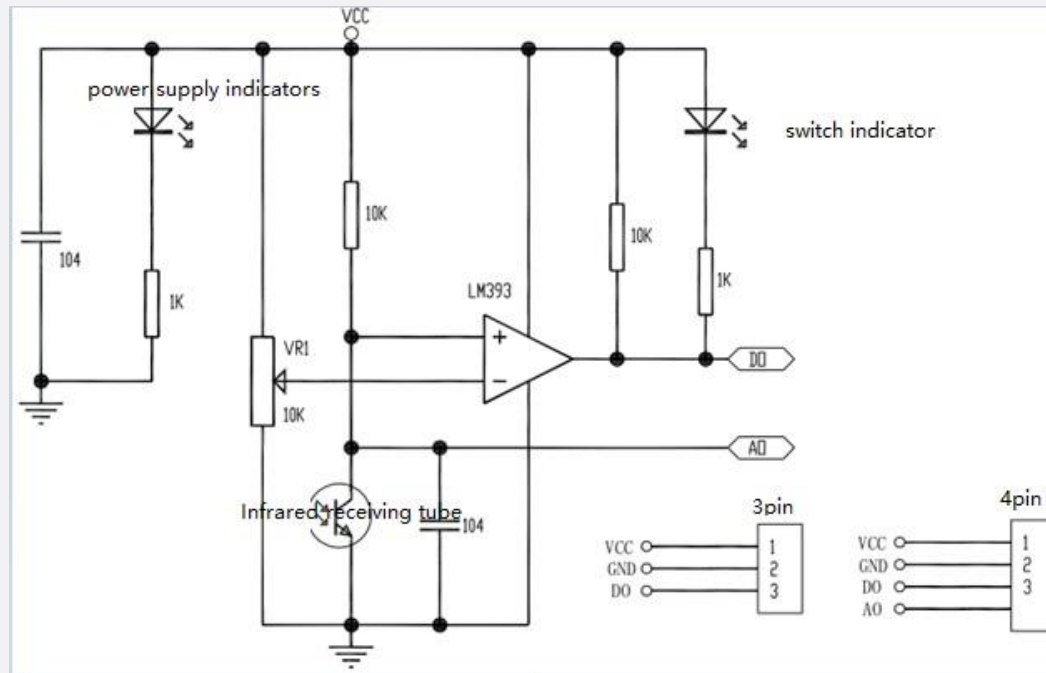
- **Arduino UNO R4/R3 Board** : Arduino can be used to develop stand – alone interactive objects or can be connected to software on your computer
- **Flame Sensor** : flame detector is a type of sensor that can detect and respond to the presence of a flame. These detectors have the ability to identify smokeless liquid and smoke that can create open fire. For example, in boiler furnaces flame detectors are widely used, as a flame detector can detect heat, smoke, and fire.
- **Red LED** : Ordinary red LEDs normally function as light emitters, but they can also function as photosensors. A single LED can even function as both a light emitter and a light detector in the same circuit
- **Buzzer** : The piezo, also known as the buzzer, is a component that is used for generating sound. It is a digital component that can be connected to digital outputs, and emits a tone when the output is HIGH.

- Type C USB Cable : allows those devices to transfer data faster while staying in a much smaller form factor. Recently, the acceptance of Type C for industrial applications, such as machine vision cameras, medical imaging sensors, 3D capturing devices, etc., has been accelerated.
- Bread board : The breadboard also called plug-in board – makes experimenting with electronic parts immensely easier. The components can simply be plugged into the breadboard without soldering them. Circuits can be plugged directly onto the breadboard.

Block Diagram : Fire Alarm By Using Sensor



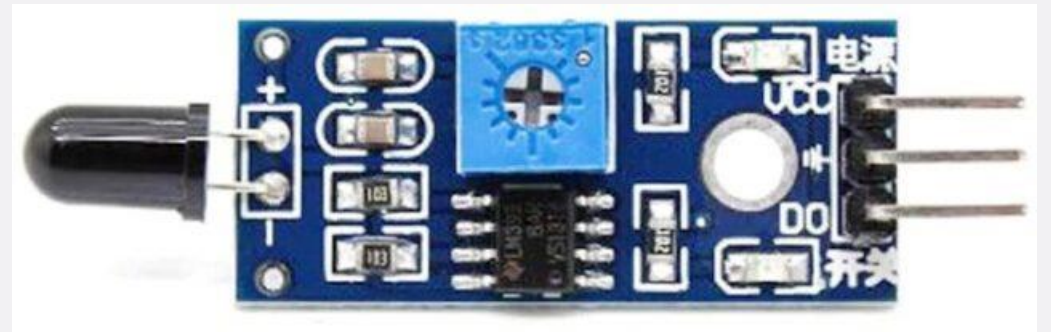
Circuit Diagram : Fire Alarm By Using Sensor



- Here is a simple circuit diagram for this project. Connect the flame sensor's VCC to Arduino's 5V and GND to Arduino's GND, and its digital output pin to Arduino's pin 4. The buzzer's positive lead connects to Arduino's pin 3, and its negative lead to another GND pin.

Flame Sensor

- Flame sensor is a device that can detect the presence of fire or other light sources within a specific wavelength range, typically from 760nm to 1100 nm.
- Flame Sensor
- It is an essential component in various applications, especially in safety systems where fire detection is crucial. The Flame Sensor is one such device that utilizes the YG1006 sensor, a high-speed and highly sensitive NPN silicon phototransistor, to detect flames.

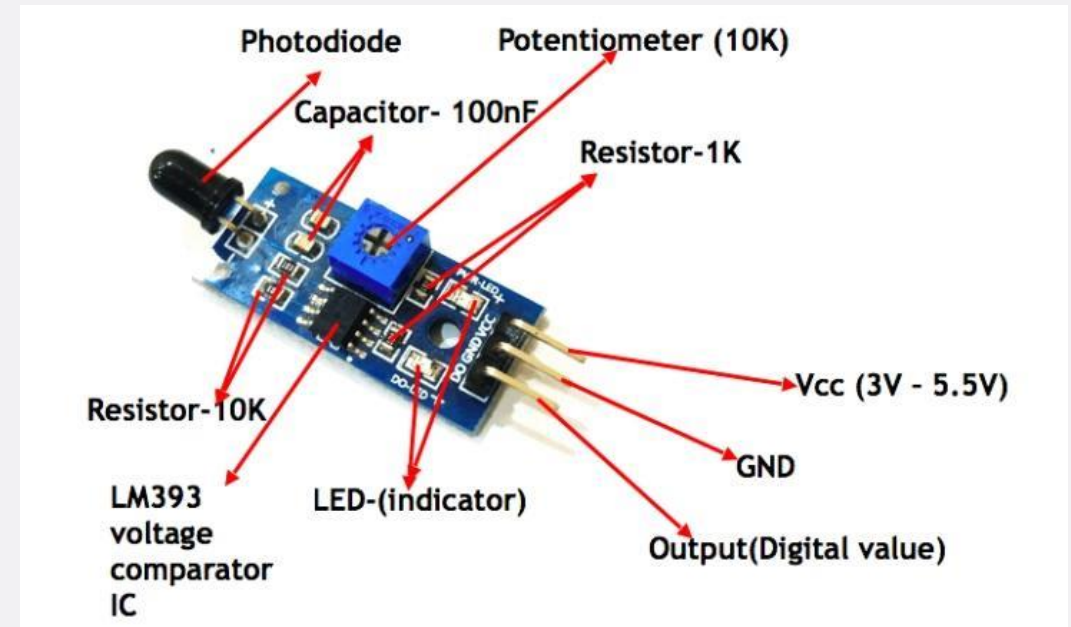


Working Of Flame Sensor

Working of the Flame Sensor

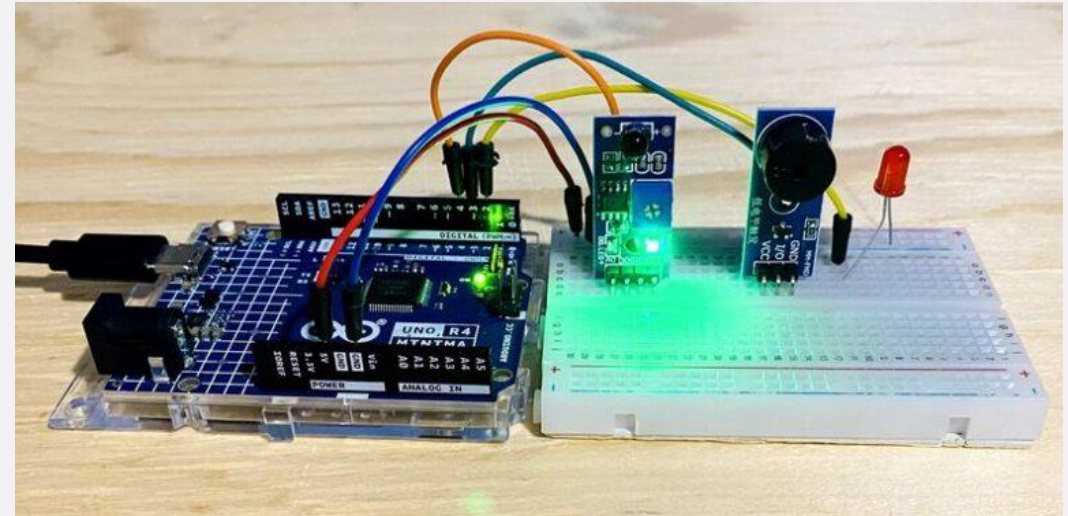
IR sensors like all other photosensor work on the principle that a photon of sufficient energy can knock out electrons so that the resistance of the circuit is changed.

An IR sensor consists of an emitter, detector, and associated circuitry. The circuit required to make an IR sensor consists of two parts; the emitter circuit and the receiver circuit.



Testing and Results

- When the flame sensor detects a flame, it sends a digital LOW signal to Digital Pin 4 on the Arduino. The code is written such that when the Arduino detects this LOW signal, it performs two actions.
- Fire Detection System Arduino
- It sends a HIGH signal to Digital Pin 2, turning on the LED. This serves as a visual indicator that a flame has been detected. It also activates the buzzer on Digital Pin 3 to play a melody.
- When the flame sensor does not detect a flame, it sends a HIGH signal to the Arduino, which results in the code sending a LOW signal to the LED pin, turning it off, and stopping any tone on the buzzer.



Application

- An early fire detection system can save lives by alerting building occupants to the presence of a fire and allowing them to evacuate safely. Early warning alerts can notify a select group of people of a potential issue in the making.
- A flame detector is a type of sensor that can detect and respond to the presence of a flame. These detectors have the ability to identify smokeless liquid and smoke that can create open fire. For example, in boiler furnaces flame detectors are widely used, as a flame detector can detect heat, smoke, and fire
- Fire Alarm Circuits are very useful in homes, offices, schools, labs, etc. To detect and prevent any disasters due to fire. Fire Alarm Systems can work as a stand – alone devices or be a part of a complex home security system with other security features like smoke detection, intruder alert, motion detection

Conclusion

Creating a fire detection system with an Arduino and a flame sensor is a great project for hobbyists and those looking to understand the basics of electronic sensors and alarms. It demonstrates the potential of simple components and how they can be used to create life-saving systems. With the right setup and programming, anyone can build a basic fire detection system that increases safety and provides a quick response to potential fire outbreaks.

Acknowledge

We Would like to express my special thanks of gratitude to our teacher Aasavari Bhalerao . Who gave us the opportunity to do this wonderful project. Of Basic Electronics Engineering on ‘Fire Alarm by using Sensor’ , Who also helped us in completing my project. We come to know about so many new things we are really thankful to them.